Previously Published - AccessRefinery: Fast Mining Concise Access Control Intents on Public Cloud

by

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(Manuscript submitted to ACM Transactions on Software Engineering and Methodology)

Dear Editor-in-Chief,

Thank you for considering our paper titled "*AccessRefinery: Fast Mining Concise Access Control Intents on Public Cloud*" for publication in the ACM Transactions on Software Engineering and Methodology (TOSEM). In this paper, we propose *AccessRefinery*, a general method for mining access control intents with high scalability and conciseness. The preliminary version of this paper has been accepted by the International Conference on the Foundations of Software Engineering (FSE) 2026 in a series organized by ACM. This submission extends the preliminary version in three key aspects:

- We observe that *AccessRefinery* can effectively reduce both the execution time and the number of intents. However, equivalence class (EC) partitioning becomes a dominant part of the total execution time, accounting for 71–98% and 16–93% of the total time for *intent mining* and *intent reduction*, respectively.
- We design *Incremental MCP* and *Pruning Reducer* to accelerate EC partitioning for *intent mining* and *intent reduction*, respectively. For *intent mining*, *Incremental MCP* incrementally partitions ECs by analyzing only the variables and values affected by a policy update, avoiding EC partitioning from scratch. For *intent reduction*, *Pruning Reducer* first identifies the necessary intents that must be included in the final intent set and then performs the minimum set-cover computation only on the remaining potentially necessary intents, thereby reducing the amount of EC partitioning required. We also formally prove the correctness of *Pruning Reducer*.
- We use experiments to illustrate three points: (a) *Incremental MCP* and *Pruning Reducer* together accelerate *AccessRefinery* by about one order of magnitude; (b) *Incremental MCP* accelerates *intent mining* by about one order of magnitude, while *Pruning Reducer* accelerates *intent reduction* by about one to two orders of magnitude; and (c) EC partitioning accounts for 71.0–98.0% and 16.4–93.2% of the total execution time in *intent mining* and *intent reduction*, respectively.

Moreover, as a minor contribution, we have open-sourced the implementations of *Incremental MCP* and *Pruning Reducer* at `https://github.com/XJTU-NetVerify/accessrefinery` to ensure that all experiments could be reproduced. Additionally, we added a synthetic dataset (marked as 05-Key*) to more comprehensively evaluate our acceleration techniques.

Thanks again for your time and effort!

Yours sincerely,

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